

The Program in Action

When you run *hacking.py*, the output will look like this:

```
Hacking Minigame, by Al Sweigart al@inventwithpython.com
Find the password in the computer's memory:
0x1150  $],>@|~~RESOLVE^      0x1250  {>+)<!?CHICKEN,%
0x1160  }@% -:;/$^(<|<|!(    0x1260  .][})?#@#ADDRESS
0x1170  _;)[#?<&~$~+&}}      0x1270  ,#=>}<{/DESPITE
0x1180  %[!]{REFUGEE@?~,      0x1280  }/.)!-DISPLAY%/
0x1190  _[^%[@]^<_+{ _$~      0x1290  =>>,:%? _?@+{##.
0x11a0  )?~/)+PENALTY?-=      0x12a0  >[,?*#IMPROVE@$/
--snip--
Enter password: (4 tries remaining)
> resolve
Access Denied (2/7 correct)
Enter password: (3 tries remaining)
> improve
A C C E S S   G R A N T E D
```

How It Works

This game has a hacking theme, but it doesn't involve any actual computer hacking. If we'd just listed the possible words on the screen, the gameplay would have been identical. However, the cosmetic additions that mimic a computer's memory banks convey an exciting feeling of computer hacking. The attention to detail and user experience turn a plain, boring game into an exciting one.

```
1. """Hacking Minigame, by Al Sweigart al@inventwithpython.com
2. The hacking mini-game from "Fallout 3". Find out which seven-letter
3. word is the password by using clues each guess gives you.
4. View this code at https://nostarch.com/big-book-small-python-projects
5. Tags: large, artistic, game, puzzle"""
6.
7. # NOTE: This program requires the sevenletterwords.txt file. You can
8. # download it from https://inventwithpython.com/sevenletterwords.txt
9.
10. import random, sys
11.
12. # Set up the constants:
13. # The garbage filler characters for the "computer memory" display.
14. GARBAGE_CHARS = '~!@#$%^&*()_+-={}|[]|;:.,.<>?/'
15.
16. # Load the WORDS list from a text file that has 7-letter words.
17. with open('sevenletterwords.txt') as wordListFile:
18.     WORDS = wordListFile.readlines()
19. for i in range(len(WORDS)):
20.     # Convert each word to uppercase and remove the trailing newline:
21.     WORDS[i] = WORDS[i].strip().upper()
22.
23.
```